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Malignancy-related mortality following kidney transplantation is common

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There is a paucity of studies describing malignancy-related mortality after kidney transplantation. To help quantify this, we extracted data for all kidney-alone transplant procedures performed in England between April 2001 and March 2012. Data linkage analysis was performed between Hospital Episode Statistics and the Office for National Statistics to identify all deaths occurring in this cohort. Among 19,103 kidney transplant procedures analyzed (median follow-up 4.4 years), 2085 deaths occurred, of which 376 (18.0%) were due to malignancy (crude mortality rate 361 malignancy-related deaths per 100,000 person-years). Common sites of malignancy-related death were lymphoma (18.4%), followed by lung (17.6%) and renal (9.8%), with 14.1% unspecified. The risk of malignancy-related death increased with age: under 50 (0.8%), 50–59 (2.5%), 60–69 (4.8%), 70–79 (6.5%) and over 80 years (9.1%). Age- and gender-stratified malignancy-related mortality risk difference was higher in the transplant compared with the general population. Cox proportional hazard models identified increased age, pretransplant history of malignancy and deceased-donor kidney transplantation to be independently associated with risk for post-transplant death from malignancy. Thus, malignancy as a cause of post-kidney transplantation death is common and requires heightened surveillance.

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Kidney transplantation is widely attributed with an increased incidence of malignancy after transplantation versus the general population,[1–3] and broadly on par with comparable immune deficiency states such as HIV/AIDS.[4] The etiology of malignancy after transplant is primarily due to the effects of immunosuppression, with suggestions of an important interplay with viral pathogens that contributes to pathophysiology.[5] By contrast, the literature on malignancy-related mortality after transplantation is scarce, with disparate mortality risk documented comparing solid organ transplant recipients to the general population.[6,7] A similar analysis in a different cohort of patients such as in the United Kingdom has not been undertaken to confirm these findings. In addition, no study has yet provided information on the site of malignancy-related mortality in kidney transplantation and whether site-specific cancer mortality can be stratified along demographic factors such as age, gender and/or ethnicity.

To inform clinical practice, we undertook a population-based cohort analysis of all deaths occurring after kidney transplantation in England over the past decade to determine causes, classifications and predictors of malignancy-related mortality. The aim of our study was to determine the overall and site-specific risk for malignancy-related mortality after kidney transplantation.

RESULTS

A total of 19,688 kidney transplant procedures in England were recorded in the Hospital Episode Statistics (HES) data for adult (18,499) and pediatric ($n = 1189$) kidney allograft recipients—excluding those with missing demographic data ($n = 585$), we had a cohort of 19,103 for further analysis (104,154 patient-years for entire study cohort). There was no missing data with regard to mortality events.

Table 1 demonstrates baseline characteristics of our study cohort. The median age for the whole cohort was 45 years (interquartile range 34–55 years). In all, 11,673 (61.1%) of the study cohort were men, with 7430 (38.9%) women. Ethnic breakdown of the study cohort comprised of white (13,695, 71.7%), Black or Black British (934, 4.9%), Asian or Asian British (1704, 8.9%), Chinese (81, 0.4%), mixed (166, 0.9%),

Table 1 | Baseline characteristics of kidney allograft recipients

Variable	Proportion (%)
Age (years)	
<50	9060 (47.6%)
50–59	4562 (23.9%)
60–69	3733 (19.5%)
70–79	1631 (8.5%)
⩾80	117 (0.6%)
Gender	
Male	11,673 (61.1%)
Female	7430 (38.9%)
Ethnicity	
White	13,695 (71.7%)
Black/Black British	934 (4.9%)
Asian/Asian British	1704 (8.9%)
Chinese	81 (0.4%)
Mixed	166 (0.9%)
Other	350 (1.8%)
Unknown	2173 (11.4%)
Socioeconomic deprivation (IMD 2010)	
1 (most deprived)	4203 (22.0%)
2	4197 (22.0%)
3	3765 (19.7%)
4	3490 (18.3%)
5 (least deprived)	3448 (18.0%)
Donor	
Living	6262 (32.8%)
Deceased	12,841 (67.2%)

Abbreviation: IMD, index of multiple deprivation.

other ethnic group (350, 1.8%), and unknown (2173, 11.4%). Socioeconomic deprivation quintiles were (from most to least deprived, respectively): 1 (4203, 22.0%), 2 (4,197, 22.0%), 3 (3765, 19.7%), 4 (3490, 18.3%), and 5 (3448, 18.0%). Living donor transplantation occurred in 6262 (32.8%) of all kidney transplant procedures reported. Diabetes mellitus classification was the most common medical comorbidity recorded in 2968 (15.5%) of all kidney allograft recipients.

Data accuracy

To verify the quality of the HES data, we extracted information from the UK National Transplant database to determine the number of transplants performed during the same time frame. During the time from April 2001 and March 2012, 19,405 kidney transplant procedures were registered (19,241 kidney alone, 49 *en bloc* kidneys and 115 double kidneys transplants). This identifies a small discrepancy of 283 kidney-alone transplant procedures that have been over-reported in the HES data in comparison to UK National Transplant Database records (concordance 98.6% between both data sets). However, it should be highlighted that HES data records 'date of admission', whereas UK National Transplant database registers 'date of transplant', which could contribute to this apparent incongruity.

Cancer mortality after kidney transplantation

By cross-referencing both HES and Office for National Statistics (ONS) databases, we identified 2085 deaths that occurred after kidney transplantation with median follow-up 4.4 years (interquartile range 2.2–7.3 years). Follow-up time consisted of 104,154 patient-years, with similar 4.4 years median follow-up in patient-years. Three hundred and seventy-six deaths were due to malignancy (2.0% of overall study cohort and 18.0% of total deaths). This equates to a crude mortality rate of 361 malignancy-related deaths per 100,000 person-years in our study cohort. Only death from infection (20.8%, 416 deaths per 100,000 person-years) and cardiovascular disease (21.2%, 424 deaths per 100,000 person-years) was more frequently documented as the leading cause of death. The most common site of malignancy-related death was lymphoma (18.4% of all cancer-related deaths), followed by lung (17.6%) and renal (9.8%), respectively (14.1% were unspecified).

Age-, sex-, and ethnicity-specific malignancy mortality rate

Table 2 highlights the risk for malignancy-related death (all-cause and location-specific) for our study cohort stratified by age, gender, and ethnicity. Age was an important and strong risk factor for cancer-related death, with increasing risk for cancer-related death as age increased: <50 (0.8%), 50–59 (2.5%), 60–69 (4.8%), 70–79 (6.5%), and over 80 (9.1%). There were only five malignancy-related deaths in the pediatric cohort (aged under 18 years) and all were classified as caused by lymphoma. Table 2 demonstrates the split in location of malignancy-related deaths into three age brackets of <50, 50–69, and over 70 years, respectively.

Lung cancer was the most common malignancy-related death in men (18.6% of all cancer deaths), with lymphoma cancer being the most common in women (18.7% of all cancer deaths). There was no significant difference in the rate of cancer-related death or location of cancer with ethnicity in our cohort, although a trend was observed toward less overall malignancy-related death in non-whites (1.3%) versus whites (2.1%) ($P=0.077$), respectively. In a comparative analysis of transplant versus general population, Figure 1 compares the age-stratified mortality risk difference between the transplant and general population and demonstrates higher malignancy-related mortality risk in the transplant cohort for the majority of age groups. Figure 2 compares gender-stratified mortality risk difference between the transplant and general cohort and again demonstrates higher malignancy-related mortality risk for both male and female kidney allograft recipients versus the general population. No comparative analysis for ethnicity was performed because of the absence of robust mortality data for England based on ethnicity.

In addition to the above demographic stratifications, we compared live-donor versus deceased-donor kidney transplant recipients for risk of malignancy-related death. We found live-donor kidney allograft recipients had a lower risk of malignancy-related death versus deceased-donor kidney allograft recipients (1.1% vs. 2.4%, respectively; $P<0.001$).

Table 2 | Incidence of malignancy-related death (and location) per 100,000 person-years stratified by age, sex, and ethnicity

Cancer	Age (years)			Sex		Ethnicity				
	<50	50–69	>70	Male	Female	White	Black	Asian	Other	Unknown
All	145.5	661.4	1396.8	380.4	330.5	387.4	265.0	253.8	314.9	329.4
Renal	10.8	68.0	179.1	39.3	29.6	36.6	22.1	23.1	35.0	42.1
Upper GI	7.7	57.2	35.8	33.0	14.8	29.8	44.2	11.5	0.0	14.0
Lower GI	6.2	57.2	179.1	33.0	22.2	27.1	0.0	11.5	35.0	56.1
Lung	17.0	133.4	214.9	70.7	51.8	65.0	22.1	23.1	70.0	91.1
Lymphoma	57.3	81.6	71.6	69.2	61.7	69.1	66.2	34.6	105.0	63.1
Breast	7.7	13.6	71.6	1.6	27.1	10.8	22.1	0.0	35.0	14.0
GU (not renal)	1.5	21.8	35.8	7.9	12.3	13.5	0.0	0.0	0.0	0.0
Prostate	0.0	8.2	107.4	9.4	0.0	8.1	0.0	0.0	0.0	0.0
Hematological	4.6	19.1	0.0	7.9	12.3	12.2	0.0	0.0	0.0	7.0
Skin	0.0	27.2	71.6	11.0	12.3	14.9	22.1	11.5	0.0	0.0
Pancreas	7.7	24.5	35.8	12.6	17.3	17.6	0.0	0.0	0.0	0.0
Liver	0.0	24.5	35.8	9.4	9.9	10.8	0.0	0.0	35.0	0.0
Female	4.6	13.6	35.8	0.0	22.2	6.8	0.0	0.0	0.0	21.0
Other[a]	1.5	21.8	35.8	12.6	4.9	12.2	0.0	0.0	0.0	7.0
Unspecified	18.6	89.8	286.5	62.9	32.1	52.8	66.2	34.6	0.0	14.0

Abbreviations: GI, gastrointestinal; GU, genitourinary.
[a]Endocrine, brain, heart, mesothelioma, and testicular.

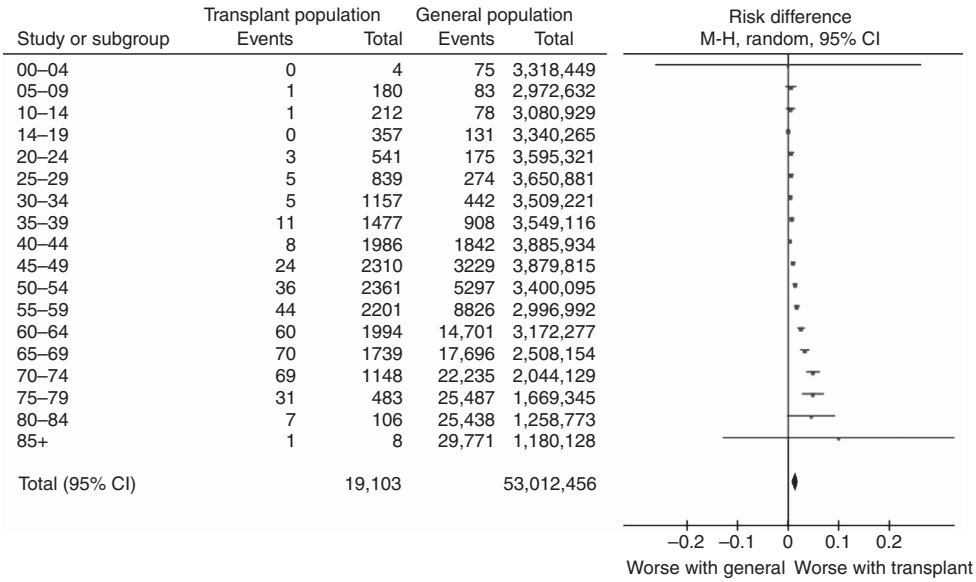


Figure 1 | Mortality risk ratio comparing kidney transplant versus general population in England (age stratified).

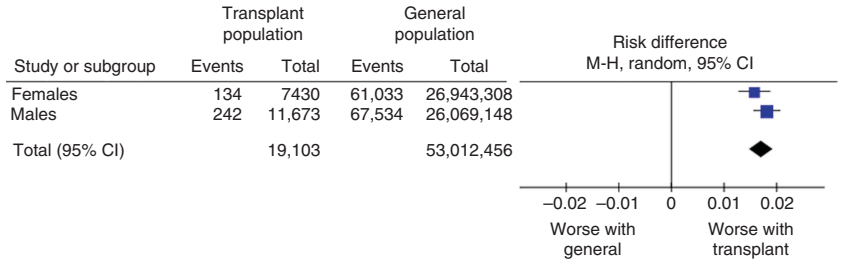


Figure 2 | Mortality risk ratio comparing kidney transplant versus general population in England (gender stratified).

No difference was observed in location of malignancy-related death between live-donor versus deceased-donor kidney recipients ($P = 0.172$).

Relative survival analysis

We undertook relative survival analysis to provide a measure of excess mortality caused by cancer in kidney allograft

recipients versus patients in the general population and the output is outlined in Table 3. Relative excess risk in kidney allograft recipients was higher with increasing age and in female subjects (borderline significance), whereas it was lower beyond the first year post-transplantation and in the more contemporary era (2006–2012 vs. 2002–2006), although the latter was of borderline significance.

Table 3 | Relative survival analysis of kidney allograft recipients versus comparative general population life tables from the Office of National Statistics

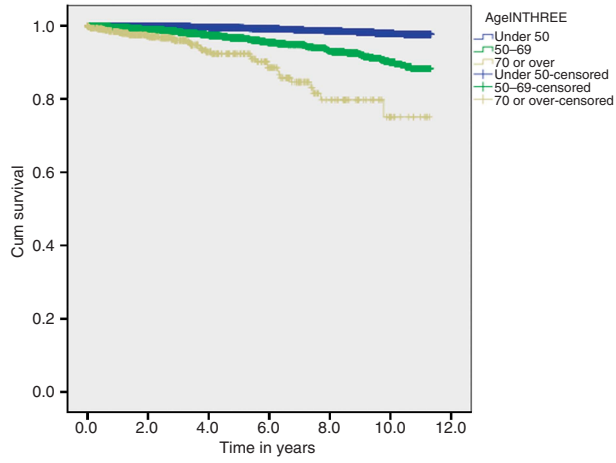
Parameter	Relative excess risk	95% Confidence interval	*P*-value
Follow-up time			
1 year	1.00	—	—
2 years	0.47	0.38–0.57	<0.001
3 years	0.47	0.38–0.58	<0.001
4 years	0.54	0.44–0.67	<0.001
5 years	0.56	0.45–0.71	<0.001
Gender			
Male	1.00	—	—
Female	1.07	0.94–1.23	0.07
Period			
2002–2006	1.00	—	—
2006–2012	0.87	0.75–1.00	0.06
Age group (years)			
0–49	1.00	—	—
50–59	2.61	2.21–3.09	<0.001
60–69	4.43	3.74–5.24	<0.001
70–79	6.85	5.20–9.03	<0.001
80 and over	4.77	0.41–55.51	0.212

Previous history of malignancy

Only 74 recipients (0.4%) in this cohort had a pretransplantation history of malignancy. Recipients with pretransplant history of cancer had higher risk of post-transplant death from malignancy compared with those without any previous history (17.6% vs. 1.9%, $P<0.001$). Recipients with pretransplant history of cancer were more likely to die from renal malignancy compared with those without pretransplant cancer history (9.5% vs. 0.2%, $P<0.001$). Indeed over half of all malignancy-related deaths in recipients with pretransplant cancer history were renal in origin (53.8%) compared with 8.3% for those with no pretransplant cancer history ($P<0.001$). We were unable to ascertain pretransplant cancer location from HES data so were unable to determine if they corresponded to post-transplant cancer death locations.

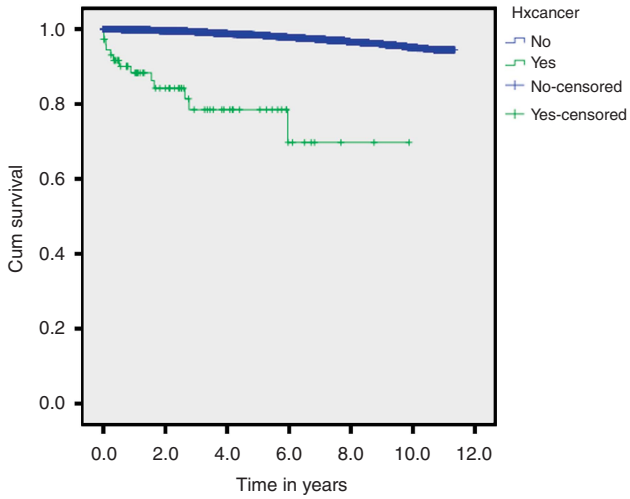
Survival and regression analysis for causes of malignancy-related mortality

Kaplan–Meier survival analysis was performed to assess unadjusted difference in risk of malignancy-related death after kidney transplantation, with Figures 3 and 4 representing survival difference between different ages and previous cancer history, respectively, among our English transplant cohort. Cox proportional hazard modeling was performed to assess the impact of competing and confounding variables upon malignancy-related mortality. Variables found to be independently associated with increased cancer mortality risk after kidney transplantation in England over the past decade included increased age, receipt of a deceased-donor kidney, and pre-transplant malignancy history (see Table 4). This remained the case in a competing risk analysis model assessing against the competing risk of other deaths versus



TIme (years)	0	1	2	3	4	5	6	7	8	9	10	11
Age < 50	11,421	10,471	9307	8106	6915	5771	4637	3616	2706	1792	982	290
Age 50–69	7037	6139	5158	4190	3336	2656	2056	1586	1126	754	381	81
Age 70+	600	479	365	258	178	142	103	65	42	24	11	3

Figure 3 | Kaplan–Meier plot of risk for malignancy-related death after kidney transplantation in England (between 2001 and 2012) stratified by age groups: (1) under 50 years; (2) 50–69 years; and (3) 70 years and over.



Time (years)	0	1	2	3	4	5	6	7	8	9	10	11
No previous cancer	19,029	17,042	14,795	12,532	10,413	8557	6793	5268	3875	2572	1377	377
Previous cancer	74	51	39	26	20	16	7	3	2	1	0	0

Figure 4 | Kaplan–Meier plot of risk for malignancy-related death after kidney transplantation in England (between 2001 and 2012) stratified by history of pretransplant cancer.

Table 4 | Cox regression proportional hazard model analysis of malignancy-related mortality after kidney transplantation

Variable	Hazard ratio	95% Confidence interval	*P*-value
Age (years)			
70+	1.000	—	—
50–69	0.415	0.292–0.590	<0.001
<50	0.084	0.057–0.125	<0.001
Medical comorbidities			
Cerebral vascular accident	2.069	1.124–3.810	0.020
Cancer	7.653	4.231–13.844	<0.001
Living donor transplant	0.702	0.537–0.918	0.010

malignancy-related deaths (see Supplementary Figure S1 online).

DISCUSSION

This study demonstrates that malignancy is a common cause of death among kidney allograft recipients, with similar incidence to cardiovascular events and infection as a cause of death after kidney transplantation. Malignancy-related mortality risk difference, stratified by both age and gender, was higher in our transplant cohort in comparison to the general population. The most frequently documented sites of malignancy on death certification were post-transplant lymphoproliferative disease, lung and renal. Pretransplant history of malignancy was a strong risk factor for malignancy-related death after transplantation, with over half of malignancy-related deaths within this cohort due to renal cancer. Finally, Cox regression analysis identified increasing age, deceased-donor kidney transplantation, and previous history of either malignancy or cerebrovascular accident as independent factors associated with risk for malignancy-related death after kidney transplantation.

The literature predominantly focuses on *incidence* of malignancy after kidney transplantation, which has been clearly demonstrated to increase after kidney transplantation for the majority of cancers (often driven by a viral etiology). Sampaio *et al.*[8] recently described the incidence of malignancies after kidney transplantation in the United States Renal Data System and cited lymphoma, lung, prostate, and kidney as the four most common malignancies (similar to mortality rates in our analysis). Our study both complements and contradicts the few publications that have explored cancer mortality rates after kidney transplantation. Kiberd *et al.*[7] identified 1937 malignancy-related death among 164,078 first kidney allograft recipients recorded in the United States Renal Data System between January 1990 and December 2004. They demonstrated no difference between the expected and observed cancer mortality rate, with equivalent standardized cancer mortality ratios with the general population. Therefore, although malignancy incidence may be increased versus the general population, actual malignancy-related mortality rate is the same. The explanation from Kiberd *et al.*[7] to explain this paradox is that there are likely to be competing risks for death for kidney allograft recipients.

By contrast our results suggest that malignancy-related mortality rates are higher among kidney allograft recipient versus the general population stratified by age and gender, in keeping with malignancy-related mortality risk reported in liver and cardiothoracic transplant recipients.[6] There are

many explanations for the differing results between the two kidney transplantation studies. The methodology and patient selection reflects different study cohorts; Kiberd and co-workers analyzed first kidney allograft recipients only within the United States (1990–2004), whereas our cohort included any kidney allograft recipient within England (2001–2012). The different time era will likely reflect different immunosuppression, postoperative management, donor/recipient selection, and so on that will make a direct comparison between the two study cohorts difficult. In addition, we would recommend cautious interpretation of our comparison between transplant versus general population malignancy-related mortality data. Making direct comparisons between the transplant and general population is flawed, as adjustments cannot be fully made for certain types of people that are over-represented in a specific cohort. For example, some of the lifestyle factors and illnesses that increase the likelihood that a person will subsequently need a transplant also increase the likelihood of dying of cancer independently of the transplant. Without the ability to adjust for these factors using further standardization, we cannot ascertain if the difference in crude cancer mortality rates are due to the prevalence of risk factors in the population groups rather than the milieu of after kidney transplantation immunosuppression.

Both Kiberd *et al.*'s and our analysis confirmed older age and previous cancer history as independent predictors of malignancy-related death. However, our results also differ with regard to the impact of previous cerebrovascular accident, with our analysis suggesting an increased risk for malignancy-related death and data from Kiberd and co-workers suggesting a decreased risk. The links between stroke and cancer are well documented in the medical literature,[9–11] with malignancy firmly established as promoting a hypercoagulation state that increases risk of future arterial and venous thromboembolic events. However, Prandoni *et al.*[12] have also demonstrated an association in the opposite direction; they observed an increased risk of clinically overt malignancy in patients previously diagnosed with symptomatic, venographically proved deep-vein thrombosis. It could therefore be speculated that a similar risk may be associated with previous arterial thromboembolic events.

We did not ascertain any significant difference in mortality rates for cancer stratified by ethnicity, although there was a trend for less cancer-related death in non-whites. Hall and co-workers[13] demonstrated that, compared with white kidney allograft recipients, black kidney allograft recipients had a lower incidence of lymphoma (adjusted incidence rate ratio 0.60, $P<0.001$) but higher incidence of kidney (adjusted incidence rate ratio 2.09, $P<0.001$) and prostate cancer (adjusted incidence rate ratio 2.14, $P<0.001$). Webster *et al.*[14] reported cancer risk was increased with white ethnicity (hazard ratio 1.36, 95% confidence interval (CI): 1.12–1.89). In our analysis, not only was no statistically significant difference in mortality rate ascertained but also the trend for mortality comparing white versus black was reversed; compared with whites, black allograft kidney recipients had higher risk for death from lymphoma (25.0% vs. 16.8%) and lower risk for death from both kidney (8.3 vs. 9.4%) and prostate (0.0 vs. 2.1%) cancer. It is difficult to explain the conflicting results between our analysis and that from Hall and co-workers, but it is possible that risk of mortality does not follow difference in incidence. For example, data from the National Cancer Intelligence Network (Cancer Research UK) demonstrated that blacks versus whites had higher incidence for prostate cancer but 1- and 3-year relative survival was better.[15] In addition, the black population in the United Kingdom is ethnically different from the black population in the United States, due to geographically different historical migration, and does not serve as a direct comparison.

Risk factors for malignancy-related mortality in our analysis are broadly similar to reported risk factors for incidence of malignancy after kidney transplantation. Webster *et al.*[14] identified increased malignancy risk after kidney transplantation with older age, previous cancer history, and white race, with decreased risk associated with diabetes-related end-stage kidney disease. The increased incidence of cancer after transplantation likely represents the biological rationale for increased cancer-related post-transplant mortality. *In vitro* studies suggest that the calcineurin inhibitors cyclosporine and tacrolimus, mainstay of contemporary immunosuppression, can promote carcinogenesis through the production of cytokines that regulate tumor growth (such as transforming growth factor-β), metastasis, and angiogenesis. However, there is a lack of clinical data to support a disparate cancer incidence between the two calcineurin inhibitors.[16] Malignancy after transplantation can also have different clinical features that may explain higher mortality. For example, central nervous system involvement with post-transplant lymphoproliferative disorders (which is frequently more aggressive and fatal) occurs in approximately 30% of cases, contrasting with 1% in the non-transplant population.[17]

The elevated risk for kidney cancer in kidney allograft recipients is well described.[1–3] It is likely that there are transplant-specific risk factors for this increased risk rather than because of the state of immunedeficiency. For example, Grulich and co-workers[4] demonstrated a significantly greater standardized incidence ratio for kidney cancer in transplant recipients (6.68, 95% CI: 5.69–8.08) versus people with HIV/AIDS (standardized incidence ratio 1.50, 95% CI: 1.23–1.83) in a meta-analysis of 12 population-based studies.[4] It was not possible from these data to determine whether the kidney cancer was from native or transplanted kidney organ, although the literature suggests that the latter is rare.[18] Increased incidence of renal malignancy is likely related to prolonged periods of dialysis and subsequent development of acquired cystic disease (with the potential for malignant transformation).[18] We can speculate that reduced cancer mortality among living kidney allograft recipients reflects increased rates of pre-emptive transplantation and decreased dialysis vintage. The absence of dialysis vintage data limited

our ability to perform a comparative analysis of kidney allograft recipients versus end-stage kidney disease patients on dialysis. This should be the subject of future analysis to explore the link between kidney cancer and dialysis vintage. Out of interest, we demonstrated that over half of malignancy-related deaths in kidney allograft recipients with previous cancer history was due to renal malignancy, but further studies with alternative registry analyses will be required to confirm this interesting observation. This also touches upon the merit of screening for malignancy in kidney allograft recipients. Wong and co-workers[19] have critically appraised cancer screening guidelines and highlight the heavy extrapolation of data regarding efficacy of cancer screening strategies in the general population to the transplant population. More targeted surveillance frameworks are required for transplant populations, although the efficacy of such strategies requires further evaluation. The American Society of Transplantation screening recommendations suggest possible pretransplant strategies to minimize the risk of post-transplant malignancies,[20] although adherence to these guidelines has been demonstrated as suboptimal.[21] Effective screening strategies after transplantation lack validation for efficacy, with direct translation of preventative strategies in the general population[22] unlikely to be valid in the high-risk transplant cohort. Further work is required in this area to achieve consensus on effective malignancy screening after transplantation. However, based on our analyses we would suggest heightened awareness of cancer risk and screening would be most appropriate for older kidney allograft recipients and especially those with pretransplant cancer histories. The exceptionally high risk of renal malignancy for kidney allograft recipients with pretransplant cancer suggests routine surveillance with post-transplant imaging studies may be beneficial.

A key strength of this study is the linking between HES and ONS data streams. The completeness and national coverage of mortality data by ONS, and the ability to perform data linkage analysis, ensures comprehensive data is generated regarding mortality numbers and death certificate registration. The use of the large HES administrative data set ensures generalizability and applicability to the wider transplant community. It also provides the power to detect small effects in meaningful outcomes that are of clinical relevance. Cancer mortality in our data is a composite outcome combining both cancer incidence and survival following a cancer diagnosis. This is strength of our analysis because it leads to a final overall assessment of the impact of cancer on the population. However, it is also a weakness because patterns in cancer mortality reflect both components and it is impossible to disentangle them. Other limitations include the absence of unmeasured and/or incompletely measured covariates that are not accounted for in the analysis. It is unlikely that statistics can ever fully adjust for all confounding factors in a cohort study and this is a limitation inherent to the very nature of observational analyses. The absence of detailed information in relation to the study cohort at baseline is a limitation of the retrospective nature of this cohort study. For example, absence of immunosuppression regimen data could have an impact on risk of cancer-related mortality (although recent data would suggest similar carcinogenicity among different immunosuppressant regimens.[16] In addition, the lack of information on dialysis histories and waitlist timings are important variables notable for their absence in our analysis as they can have important influences on our end points (e.g. kidney cancer mortality). The ability to link data streams with other national data registries in the United Kingdom to obtain missing data such as dialysis vintage and allograft failure would strengthen the legitimacy and generalizability of such clinical research output and should be actively pursued. It is also difficult to ascertain from analyses such as this whether malignancy was directly *responsible* for death or simply *associated* with death. We have also previously commented upon the limitations of direct comparisons between cancer-related mortality rates between transplant versus general population in our data set. Finally registry data have well-documented limitations[23,24] that lead to inherent problems with such analyses, and acknowledgement of these shortcomings is essential in the correct interpretation of our results.

In conclusion, we have demonstrated malignancy to be a common cause of death after kidney transplantation and associated with expected (age, pretransplant history of malignancy) and unexpected (deceased-donor kidney transplantation) independent risk factors. Stratification by age and gender would suggest elevated risk of malignancy-related death versus the general population, although a direct comparison is fraught with limitations. Our contrasting results to previous studies reinforce the need for further work in this area to clarify accurately the burden of malignancy after kidney transplantation. A greater understanding of malignancy-related incidence and/or mortality risk after kidney transplantation will allow clinicians to tailor modifiable risk factors such as immunosuppression appropriately. Malignancy is a significant cause of death after kidney transplantation and requires targeted surveillance by transplant clinicians.

MATERIALS AND METHODS

Study design

This was a retrospective observational cohort study, performed on prospectively registered national data sets. This study included all kidney transplant procedures (Office of Population Censuses and Surveys Classification of Interventions and Procedures, 4th revision (OPCS-4) codes; M011–M015 and M018–M019) performed at any English kidney transplant center (adult and pediatric) between April 2001 and March 2012. Combined organ transplants (kidney with another organ) were excluded from the analysis ($n = 1255$). During this time period, 19,688 kidney transplant procedures were recorded in the HES data (635 recipients had repeat transplant performed during the 12-year period but only original transplant factored into Cox regression analysis). Five hundred and eighty-five were excluded owing to the lack of completeness in relation to

demographic information (age, gender, or socioeconomic deprivation), leaving a final cohort of 19,103 recipients for analysis.

Data sources

Data were obtained from HES,[25] an administrative data warehouse containing admissions to all National Health Service (NHS) hospitals in England. HES has details of all NHS in-patient treatment, outpatient appointments, and Emergency Department attendances in England. It includes private patients treated in NHS hospitals, patients resident outside of England, and care delivered by treatment centers (including those in the independent sector) funded by the NHS. Each HES record contains a wide range of information about an individual patient admitted to an NHS hospital, including: (1) clinical information about diagnoses and operations; (2) information about the patient, such as age group, gender, and ethnicity; (3) administrative information, such as dates and methods of admission and discharge; and (4) geographical information such as where patients are treated and the area where they live. It contains detailed records relating to individual patient treatments, with data extraction facilitated utilizing codes on procedural classifications (OPCS-4)[26] and medical classifications (World Health Organization International Classification of Disease, 10th revision (ICD-10)).[27]

With regard to outcome analysis, HES data alone has the limitation of only capturing deaths occurring in a hospital setting. To obtain the complete mortality list, the study cohort was cross-referenced with mortality data from the Office for National Statistics, which collects information on all registered deaths in the United Kingdom. Data linkage was performed by the Health and Social Care Information Centre (HSCIC), with the linkage process consisting of assigning a unique HES identifier (the HESID index) to the ONS death record. ONS mortality matching to an HESID is based on eight different criteria, with the best quality match selected and data merging proceeds. Further information regarding the data linkage process is available on the HSCIC website.[28] Combining sources this data linkage process is robust and creates a complete and comprehensive data set with regard to cancer-related mortality, which was the end point of interest in this analysis.

Study variables

Data were extracted from each kidney transplant procedure performed in England in relation to patient demographics at the time of transplant; age, gender, donor type (living vs. deceased), ethnicity (white, Black or Black British, Asian or Asian British, Chinese, mixed, other, and unknown), medical comorbidities (acute myocardial infarct, congestive heart failure, peripheral vascular disease, cerebral vascular accident, pulmonary disease, connective tissue disorder, peptic ulcer, cancer, liver disease, diabetes mellitus), and socioeconomic deprivation (based on Index of Multiple Deprivation (IMD) 2010).[29] IMD 2010 is a socioeconomic deprivation model calculated at the local level area. The model is based on the assessment of distinct domains of deprivation that can be individually measured and is composed of: (1) income deprivation; (2) employment deprivation; (3) health deprivation and disability; (4) education skills and training deprivation; (5) barriers to housing and services; (6) living environment deprivation; and (7) crime. Individual domains are measured in isolation and subsequently combined (utilizing appropriately judged weighting) into a single composite termed the IMD. On the IMD quintile scale, 1 represented the most deprived and 5 the least deprived area, respectively.

Study outcomes were to be analyzed after stratification by age, sex, and ethnicity. This study did not require institutional review board approval due to the pseudoanonymized nature of the data retrieved. This observational study was registered with a clinical trials registry (NCT01798524).

Data quality

As an observational study, we adhered to the principles of the STROBE statement and have reported this article in keeping with the recommended guidance.[30] In addition, we checked data accuracy from the HES database regarding transplant activity by corroborating HES data with the UK Transplant National Transplant database (where all transplant activity must be mandatorily reported). Transplant activity over the same dates as the study period was crosschecked between both databases to ascertain any discrepancy between data sets.

Classification of death

Owing to the subjective nature of death classification and heterogeneity of terms utilized in the death certificates, we classified all deaths into broad systems-based categories (including malignancy). Two authors (DF, JC) converted all causes of death from ONS data from string into numerical data, with a third author independently verifying the malignancy data and classifying location (SM). All final data was reviewed by AS and any discrepancy regarding classifications resolved by discussion.

Statistical methods

The primary study outcome was malignancy-related death after kidney transplantation. SPSS (Version 20.0, Chicago, IL) was utilized for data analysis, unless otherwise stated. Normality of data was assessed using the Kolmogorov–Smirnov tests. Descriptive statistics were used to estimate the frequencies. Categorical variables are presented as number (%) and continuous variables as mean ($\pm$ s.d.) or median ($\pm$ interquartile range) dependent on normality of distribution. Difference between groups was assessed with χ^2 or two-sided Fisher's exact test for categorical variables and *t*-test or Mann–Whitney *U*-test to compare continuous variables. A *P*-value <0.05 and 0.001 in the statistical analysis was considered significant and highly significant, respectively.

Cox's proportional hazards model in STATA (command stcox) was utilized. The proportionality assumption was checked for each variable and the whole model—for the main analysis, the proportionality assumption is true for all variables except cancer. Variables included in the model were age, gender, donor type (living vs. deceased), socioeconomic deprivation, ethnicity, year of transplant, and selected medical comorbidities (history of myocardial infarction, peripheral vascular disease, cerebrovascular disease, congestive cardiac failure pulmonary disease, liver disease, peptic ulcer, previous cancer, and diabetes). Review Manager 5 was utilized to generate risk difference Forest plots to compare malignancy death rates between transplant versus general populations. Relative survival analysis was performed against life tables published by the Office of National Statistics. Data were available for 2001–2011 (no 2012 data currently available) and therefore 2012 values were utilized as a proxy for 2012. Stata/IC 12 was utilized for this analysis and excess mortality was modeled using a full likelihood approach,[31] implemented by Dickman *et al.*[32]

With the assumption that data was missing at random, we simply performed list-wise deletion and excluded the missing values from the analysis. Other missing data (e.g. ethnicity) was adjusted for as dummy variables in the models as required.

DISCLOSURE

All the authors declared no competing interests.

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SUPPLEMENTARY MATERIAL

Figure S1. Output of competing risk regression model.
Supplementary material is linked to the online version of the paper at http://www.nature.com/ki

REFERENCES

1. Kasiske BL, Snyder JJ, Gilbertson DL *et al.* Cancer after kidney transplantation in the United States. *Am J Transplant* 2004; **4**: 905–913.
2. Vajdic CM, McDonald SP, McCredie MRE *et al.* Cancer incidence before and after kidney transplantation. *JAMA* 2006; **296**: 2823–2831.
3. Kyllonen L, Salmela K, Pukkala E. Cancer incidence in a kidney-transplanted population. *Transpl Int* 2000; **13**(Suppl 1): S394–S398.
4. Grulich AE, van Leeuwen MT, Falster MO *et al.* Incidence of cancers in people with HIV/AIDS compared with immunosuppressed transplant recipients: a meta-analysis. *Lancet* 2007; **370**: 59–67.
5. Morath C, Mueller M, Goldschmidt H *et al.* Malignancy in renal transplantation. *J Am Soc Nephrol* 2004; **15**: 1582–1588.
6. Na R, Grulich AE, Meagher NS *et al. De novo* cancer-related death in australian liver and cardiothoracic transplant recipients. *Am J Transplant* 2013; **13**: 1296–1304.
7. Kiberd BA, Rose C, Gill JS. Cancer mortality in kidney transplantation. *Am J Transplant* 2009; **9**: 1868–1875.
8. Sampaio MS, Cho YW, Qazi Y *et al.* Posttransplant malignancies in solid organ adult recipients: an analysis of the US National Transplant Database. *Transplantation* 2012; **94**: 990–998.
9. Grisold W, Oberndorfer S, Struhal W. Stroke and cancer: a review. *Acta Neurol Scand* 2009; **119**: 1–16.
10. Lee AY, Levine MN. Venous thromboembolism and cancer: risks and outcomes. *Circulation* 2003; **107**: 117–121.
11. Schwarzbach CJ, Schaefer A, Ebert A *et al.* Stroke and cancer: the importance of cancer-associated hypercoagulation as a possible stroke etiology. *Stroke* 2012; **43**: 3029–3034.
12. Prandoni P, Lensing AWA, Buller HR *et al.* Deep-vein thrombosis and the incidence of subsequent symptomatic cancer. *N Engl J Med* 1992; **327**: 1128–1133.
13. Hall EC, Segev DL, Engels EA. Racial/ethnic differences in cancer risk after kidney transplantation. *Am J Transplant* 2013; **13**: 714–720.
14. Webster AC, Craig JC, Simpson JM *et al.* Identifying high risk groups and quantifying absolute risk of cancer after kidney transplantation: a cohort study of 15183 recipients. *Am J Transplant* 2007; **7**: 2140–2151.
15. Cancer Incidence and Survival by Major Ethnic Group, England, 2002–2006. National Cancer Intelligence Network. Available at: http://publications.cancerresearchuk.org/downloads/product/CS_REPORT_INCSURV_ETHNIC.pdf (accessed 14 August 2013).
16. Gallagher MP, Kelly PJ, Jardine M *et al.* Long-term cancer risk of immunosuppressive regimens after kidney transplantation. *J Am Soc Nephrol* 2010; **21**: 852–858.
17. Taylor AL, Marcus R, Bradley JA. Post-transplant lymphoproliferative disorders (PTLD) after solid organ transplantation. *Crit Rev Oncol/Haematol* 2005; **56**: 155–167.
18. Muruve NA, Shoskes DA. Genitourinary malignancies in solid organ transplant recipients. *Transplantation* 2005; **80**: 709.
19. Wong G, Chapman JR, Craig JC. Cancer screening in renal transplant recipients: What is the evidence? *Clin J Am Soc Nephrol* 2008; **3**: S87–S100.
20. Kasiske BL, Cangro CB, Hariharan S *et al.* The evaluation of renal transplant candidates: clinical practice guidelines. Recommendations for outpatient surveillance of renal transplantation. *Am J Transplant* 2001; **1**(Suppl 2): 5–95.
21. Danovitch GM, Hariharan S, Pirsch JD *et al.* Clinical Practice Guidelines Committee of the American Society of Transplantation. Management of the waiting list for cadaveric kidney transplants: report of a survey and recommendations by the Clinical Practice Guidelines Committee of the American Society of Transplantation. *J Am Soc Nephrol* 2002; **12**: 528–553.
22. US Preventive Services Task Force. Available at: http://www.uspreventive servicestaskforce.org/uspstopics.htm (accessed 12 July 2013).
23. Arts DGT, de Keizer NF, Scheffer GJ. Defining and improving data quality in medical registries: a literature review, case study, and generic framework. *J Am Med Inform Assoc* 2002; **9**: 600–611.
24. Byar DP. Why data bases should not replace randomized clinical trials. *Biometrics* 1980; **36**: 337–342.
25. Hospital Episode Statistics online. Available at: www.hesonline.nhs.uk (accessed 12 January 2013).
26. OPCS-4 Classification—NHS Connecting for Health. Available at: www.connectingforhealth.nhs.uk (accessed 12 January 2013).
27. World Health Organization: International Classification of Diseases (ICD). Available at: www.who.int/classifications/icd/en (accessed 5 January 2013).
28. Health & Social Care Information Centre. A Guide to Linked Mortality Data from Hospital Episode Statistics and the Office for National Statistics. Available at: www.hscic.gov.uk/media/11668/HES-ONS-Mortality-Data-Guide/pdf/guide_to_linked_ONS_HES_mortality_V4_040613.pdf (accessed 25 July 2013).
29. Department for Communities and Local Government. The English Indices of Deprivation 2010. Available at: https://www.gov.uk/government/publications/english-indices-of-deprivation-2010 (accessed 17 January 2013).
30. von Elm E, Altman DG, Egger M *et al.* STROBE Initiative. The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement: guidelines for reporting observational studies. *Lancet* 2007; **370**: 1453–1457.
31. Esteve J, Benhamou E, Croasdale M *et al.* Relative survival and the estimation of net survival: elements for further discussion. *Stat Med* 1990; **9**: 529–538.
32. Dickman PW, Coviello E, Hills M. Estimating and modelling relative survival. *Stata J* 2013; **vv**: 1–25.